High Proxy Risk

PRD_FOREIGN_BORROWER. This is especially sensitive because it may proxy for national origin, immigration status, language background, or other protected traits.

The model may learn: Foreign borrower → higher bad rate

Instead of learning: Short Canadian credit history

POD_MARKET_CODE. Market segmentation codes are often highly correlated with neighborhood composition, regional income, and borrower demographics.

The model can learn: Certain neighborhoods = higher risk

Instead of: Affordability differences

POD_SOURCE_OF_BUSS_TYPE. Broker, retail, and mortgage specialist channels can serve different populations, so this may capture selection effects that are not purely risk-based.

The model could learn: Broker → risky

Instead of: Borrower profile → risky

Medium Proxy Risk

CRD_MONTHS_TU. It can proxy with age, new-to-Canada status, recent immigration, young borrowers.

The model could learn: 24 months TU history → higher risk

instead of: Young borrowers → higher risk

So CRD_MONTHS_TU can indirectly carry age information.

PRK_OCCUPANCY. Primary, secondary, and investment occupancy can proxy for wealth, housing stability, and household structure.

Investment properties often correlate with wealth/income, which has demographic disparities.

The model might learn: Rental property → risky

Instead of: Certain investor demographics → risky.

PRK_TENURE. may correlate with property value, location, and buyer demographics.

Vancouver downtown → Leasehold
Toronto suburbs → Freehold

The model could learn: Leasehold → risky

Instead of: Certain regional socioeconomic factors.

PRK_PROPERTY_TYPE. can also act as proxies because they reflect housing stock and neighborhood characteristics.

Downtown Toronto → Condo
Suburbs → Detached

The model could learn: Condo → higher risk

But the true underlying driver may be: Younger first-time buyers with lower savings.

PRK_CONS_TYPE. New construction often correlates with higher-income or specific demographic areas.

The model may learn: New construction → lower risk

Instead of: Higher-income borrowers → lower risk

Resale homes may be associated with: Older borrowers, Suburban markets

The model could learn: Resale → lower risk (without understanding the underlying reason)

Toronto downtown → mostly new condos
Small towns → mostly resale detached homes

Now the model is indirectly learning: Region

Instead of: Borrower risk

RURAL_IND. Rural or non-rural status can act as a geographic proxy, and geography often correlates with socioeconomic and demographic patterns.

BORYREMP. Can act as a weak proxy for age.